

Google Calendar + Gmail Integration Guide

WilliamOS (Mini-jarv) — setup, usage, and limitations

Overview

The Google integration pulls **calendar events** and **unread Gmail** into your **Inbox** as signals. You can then apply AI suggestions to create tasks, assets, and more.

Stack: Google Calendar API + Gmail API → FastAPI (`google_service.py`) → Inbox (`capture_inbox_entry`)

What's Already Built

The full OAuth and sync path is wired in the app:

Layer	Status	Location
OAuth flow	Done	<code>app/services/google_service.py</code>
Token refresh	Done	Auto-refreshes before sync
Calendar fetch	Done	Next 7 days, max 15 events
Gmail fetch	Done	Last 8 unread messages
API routes	Done	<code>app/api/routes/integrations.py</code>
UI	Done	<code>/integrations + /integrations/callback</code>
DB table	Done	<code>user_integrations</code> (+ migration for google provider)

OAuth flow

1. **Integrasjoner** → **Koble til** → POST `/integrations/google/connect`
2. Redirect to Google login
3. Google redirects to `/integrations/callback?code=...&state=...`
4. Callback calls POST `/integrations/google/complete`
5. Tokens stored in `user_integrations`

Sync flow

1. **Synk til Inbox** → POST /integrations/google/sync
 2. Fetches calendar + unread mail via Google APIs
 3. Creates Inbox lines like:
 - Google kalender: Møte (2026-08-18 10:00)
 - Google e-post: Forsikring fornyes
 4. Each line gets LLM/rule-based suggestions — apply them manually in Inbox
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What You Need to Do

1. Run Supabase migrations

In Supabase SQL editor, ensure these are applied:

```
migrations/2026-08-17_finance_health_integrations.sql
migrations/2026-08-18_google_integration.sql
```

The second migration adds google as a valid provider in user_integrations.

2. Set up Google Cloud Console (~15 min)

Go to: <https://console.cloud.google.com>

1. **Create a project** (or select existing)
2. **APIs & Services** → **Library** — enable:
 - **Google Calendar API**
 - **Gmail API**
3. **OAuth consent screen**
 - User type: **External** (for personal Gmail) or Internal (Workspace)
 - Add your email as a **Test user** while in testing mode
 - Scopes: Calendar read-only + Gmail read-only
4. **Credentials** → **Create credentials** → **OAuth client ID**
 - Application type: **Web application**
 - **Authorized redirect URIs:**
 - Local: `http://localhost:3000/integrations/callback`
 - iPhone via ngrok: `https://YOUR-SUBDOMAIN.ngrok-free.app/integrations/callback`
5. Copy **Client ID** and **Client secret**

Note: Google Cloud OAuth is free for personal/dev use. No billing required for API access at MVP scale.

3. Set environment variables

Add to your root .env:

```
GOOGLE_CLIENT_ID=your-client-id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=your-client-secret
GOOGLE_REDIRECT_URI=http://localhost:3000/integrations/callback
FRONTEND_URL=http://localhost:3000
```

Critical: GOOGLE_REDIRECT_URI must match Google Cloud **exactly** (including path /integrations/callback).

4. Restart the backend

FastAPI reads env at startup. Without restart, Integrasjoner shows "Sett GOOGLE_CLIENT_ID..." and **Koble til** stays disabled.

5. Connect in the app

1. Start backend (:8000) + frontend (:3000)
2. Log in to Mini-jarv
3. Go to **Mer** → **Integrasjoner**
4. Click **Koble til** on Google Calendar & Gmail
5. Approve Google permissions (Calendar + Gmail read)
6. Click **Synk til Inbox** → check **Inbox**

6. Use a Google account with calendar and mail

Works with personal @gmail.com and Google Workspace accounts.

Required Scopes

```
https://www.googleapis.com/auth/calendar.readonly
https://www.googleapis.com/auth/gmail.readonly
```

Defined in app/services/google_service.py. OAuth uses access_type=offline and prompt=consent so a refresh token is issued on first connect.

Known Limitations (Today)

Limitation	Detail
Manual sync only	No cron, no sync-on-login
Duplicates	Re-syncing creates duplicate Inbox items
Unread email only	Read mail is skipped

Limitation	Detail
Calendar window	7 days ahead, max 15 events
No calendar UI	Events only appear as Inbox text lines
Testing mode	Only test users you add can connect until app is verified
No auto-tasks	Sync creates Inbox signals; tasks are not auto-created

Troubleshooting

Problem	Solution
"Sett GOOGLE_CLIENT_ID..." in UI	Env vars missing, or backend not restarted
Redirect URI mismatch	Google Cloud URI must equal GOOGLE_REDIRECT_URI exactly
"Ugyldig OAuth-state"	Click Koble til again
Koble til button disabled	configured: false — check env vars
"Access blocked" / 403	Add your Gmail as test user on OAuth consent screen
ngrok on iPhone	Update both Google redirect URI and GOOGLE_REDIRECT_URI
No inbox after sync	Calendar empty next 7 days, or no unread mail
Token expired / error	Disconnect and reconnect Google

API Reference

All endpoints require authenticated user (Bearer JWT).

GET /integrations	List integration statuses
POST /integrations/google/connect	Start OAuth, returns auth_url
POST /integrations/google/complete	Complete OAuth with code + state
POST /integrations/google/sync	Sync calendar + mail to Inbox
POST /integrations/google/disconnect	Disconnect and clear tokens

Pre-Flight Checklist

- Migrations `finance_health_integrations` + `google_integration` run in Supabase
 - Google Calendar API and Gmail API enabled
 - OAuth client created with correct redirect URI
 - Your email added as test user (if app is in Testing)
 - `GOOGLE_CLIENT_ID`, `GOOGLE_CLIENT_SECRET`, `GOOGLE_REDIRECT_URI` in `.env`
 - Backend restarted after env changes
 - Logged into Mini-jarv
 - Google account has upcoming events and/or unread mail
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Future Improvements (Not Built Yet)

- Dedup on sync (avoid duplicate Inbox rows)
 - Calendar event → auto-suggest/create task
 - Email keyword parsing (`forsikring`, `faktura` → `asset/task`)
 - Auto-sync on login or scheduled daily sync
 - Dedicated calendar view in the app
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Generated for WilliamOS / Mini-jarv — see also `docs/GETTING-STARTED.md`